

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	01 Dec 2015	First Release	ONG KEAN SHENG
01	05 Oct 2016	Phrase corrective	ONG KEAN SHENG
02	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

Content

PSP 2 Rear MODULE OFFLINE ADJUSTMENT.....	3
---	---

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

PSP 2 Rear MODULE OFFLINE ADJUSTMENT

1. Adjust the DIP switch at Rear Projector Module as below Figure 1.

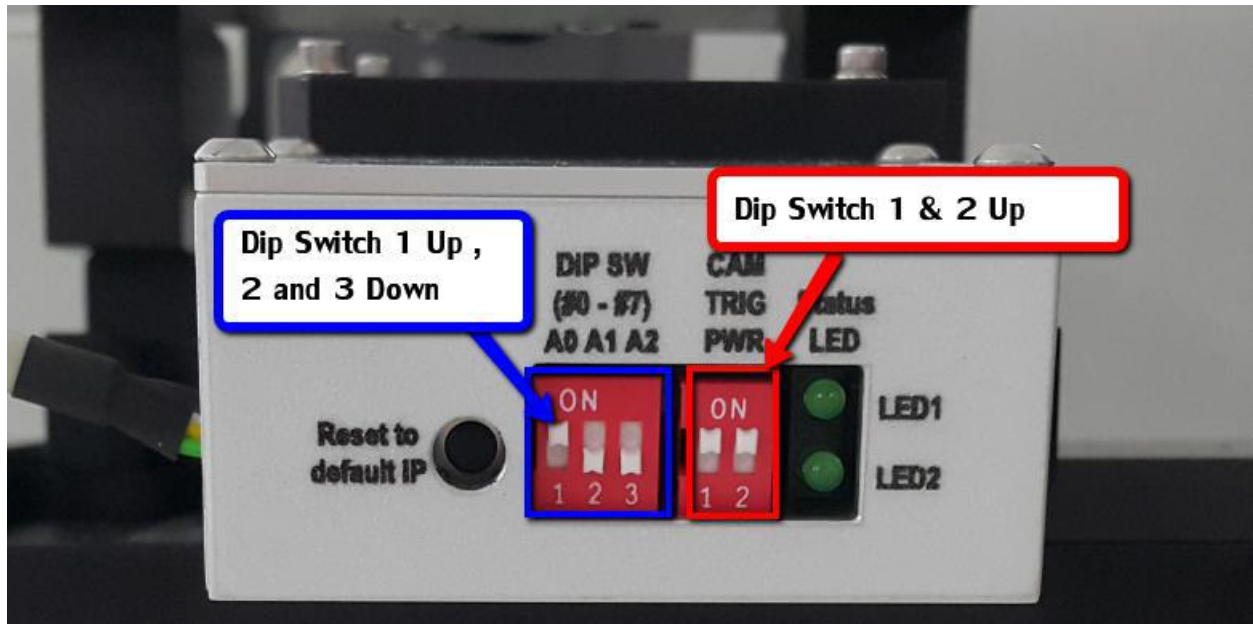


Figure 1 Rear Projector IP DIP SWITCH CONFIGURATION

2. Assemble the PSP 2 Module at "STD/XXL PSP Adjustment Offline Jig" by using M5 X 12 as below Figure 2.

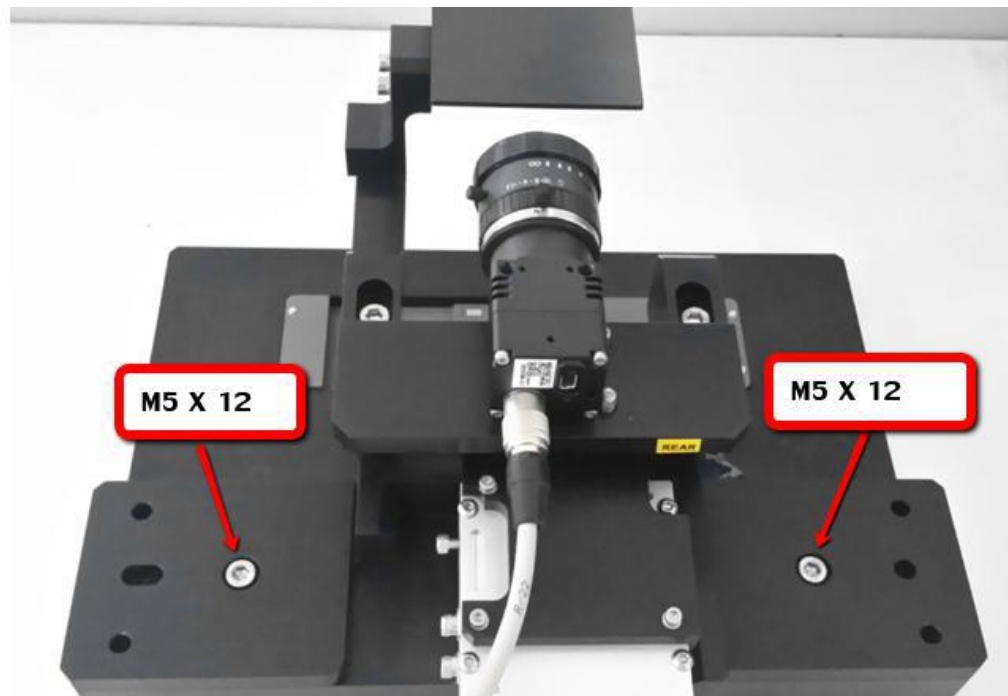


Figure 2 OFFLINE JIG ASSEMBLY

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

3. Open Front Access Door , gently allocate “STD/XXL PSP Offline Adjustment Jig ” (Part ID- 4XASPP-A002) at Granite area (May refer to below Figure 3 for the desire location)

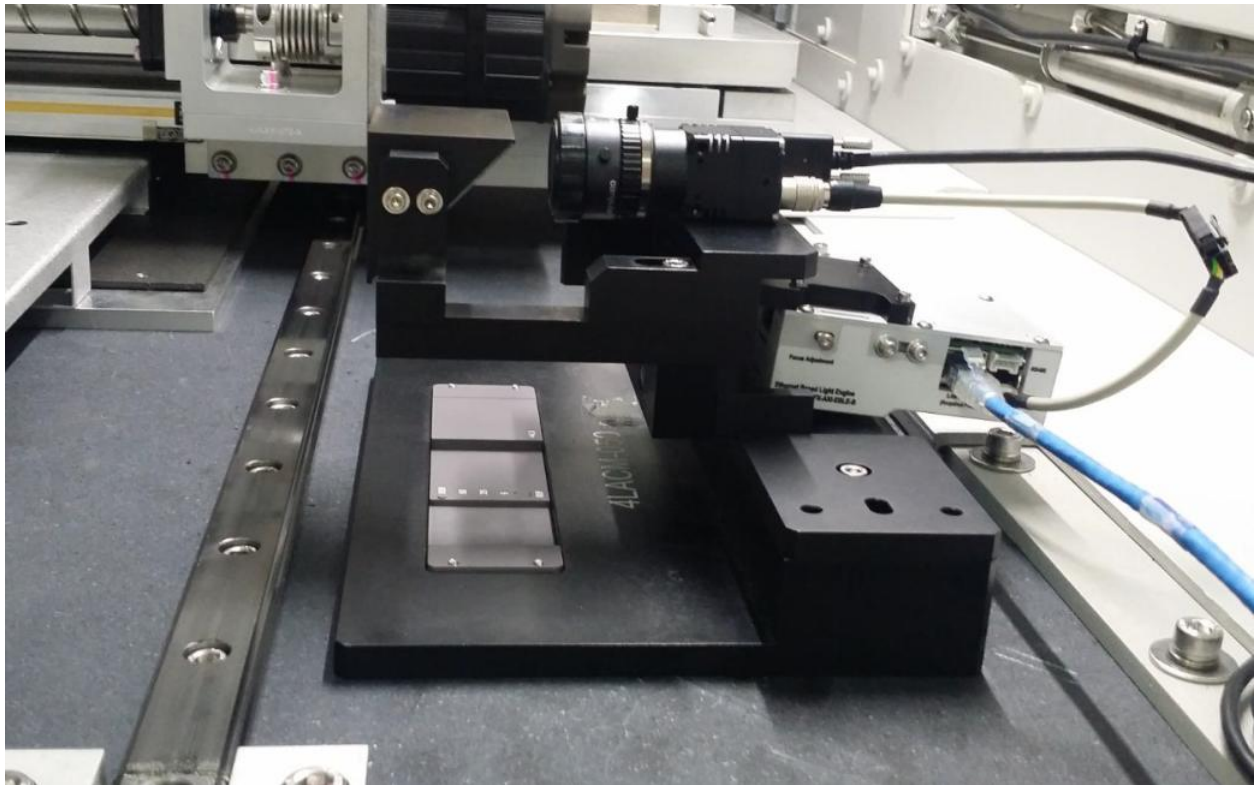


Figure 3 OFFLINE JIG & MODULE LOCATION

4. Turn Off 3 PIN PLUG Switch as below Figure 4. (Remarks : Power supply for POE Switch)

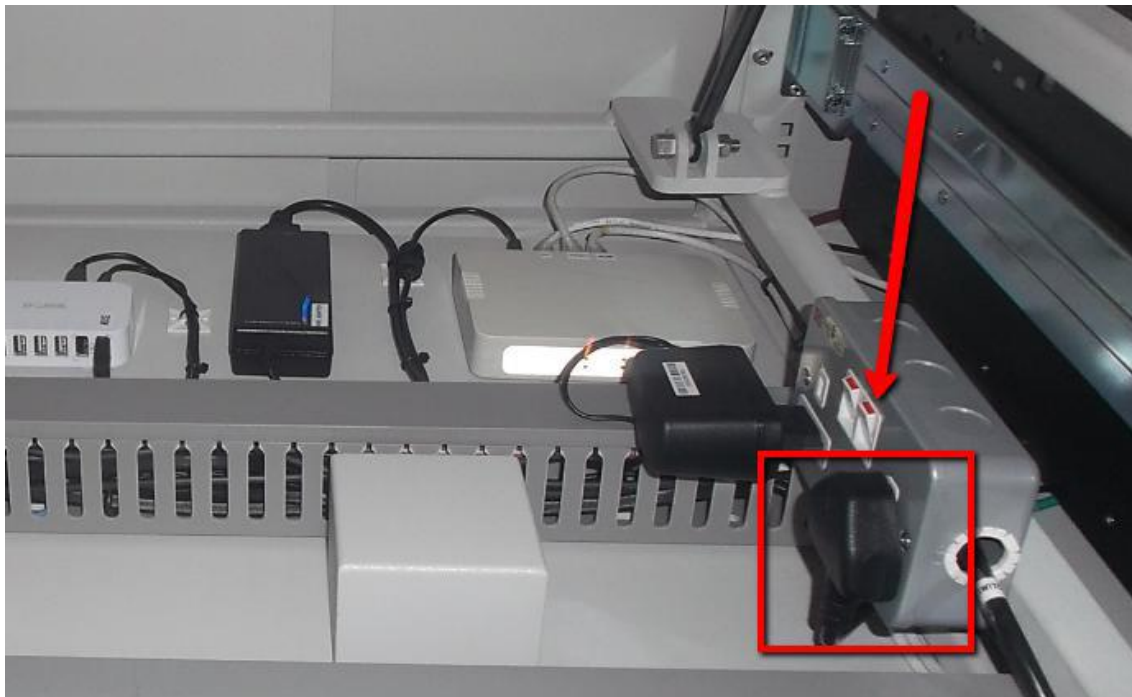


Figure 4 3 PIN PLUG POWER SWITCH

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

- Perform Marking at below cable and unplug 1 x REAR Cat 5 cable at POE SWITCH (which allocated at rear server area) as below Figure 5.

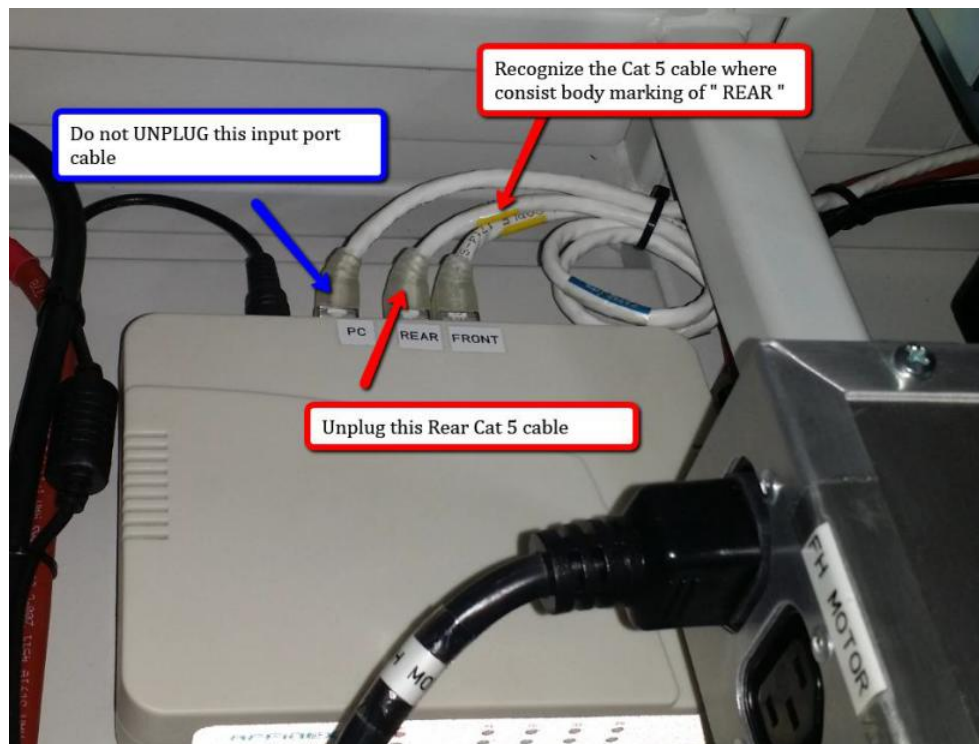


Figure 5 POE SWITCH CAT 5 CABLE

- Using another 1 x 2530-0063 (30 Feet Cat cable) and plug at POE Switch's "Rear Labeling Port" as below Figure 6.



Figure 6 Rear labeling Cat 5 Port

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

7. The other end of Cat 5 cable will be connecting to PSP 2 Projector Port as below Figure 7.

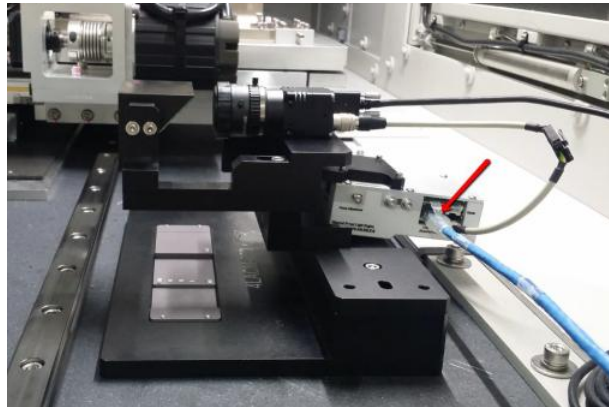


Figure 7 PSP 2 PROJECTOR PORT

8. Using 1 x 2530-0110 (Camera Cable_Area Scan_5Meter) and connect to PSP 2 Camera as below Figure 8.

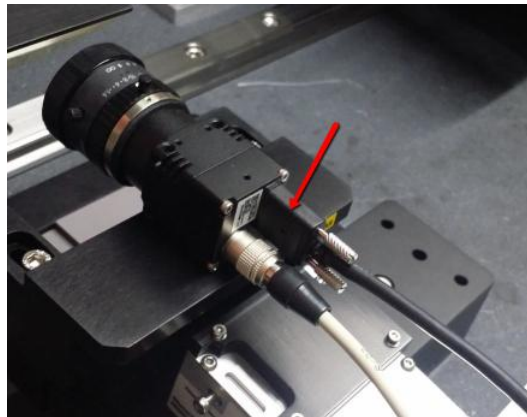


Figure 8 PSP 2 Camera Cable

Remarks: Due to Camera Cable Length, rear door may require to open to prolong the cable routing to master controller server. May refer to Figure 9.

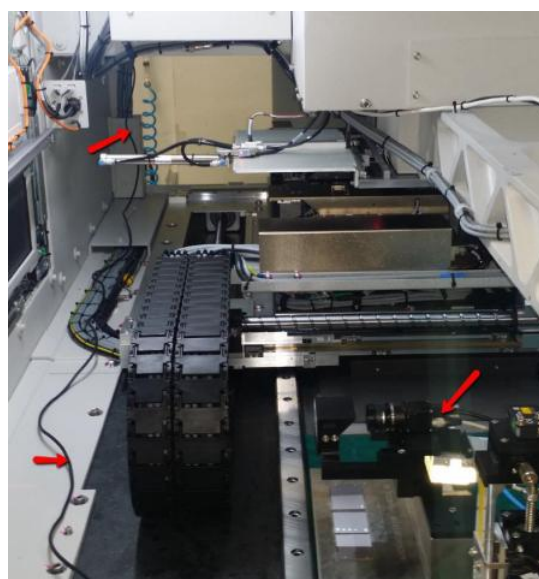


Figure 9 Camera Cable Routing (Reference Only)

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

9. The other end of the PSP 2 CAMERA CABLE may connect to Master Controller (USB PORT 3.0 Blue Color) as below Figure 10.

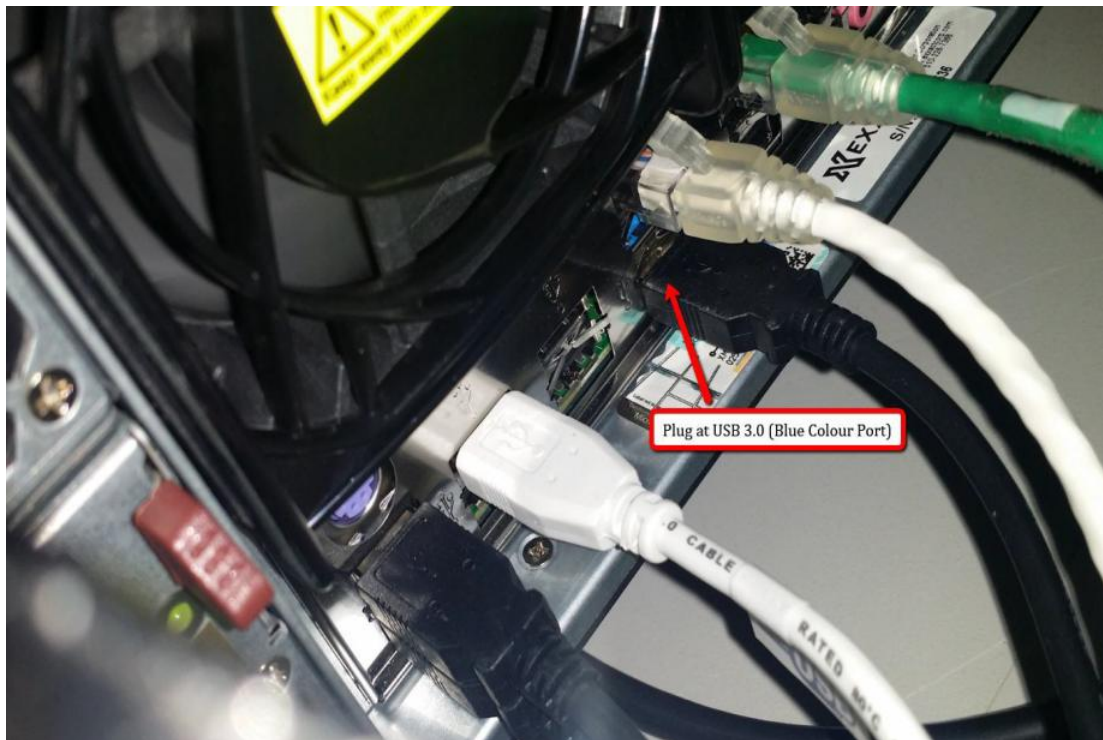


Figure 10 Rear side of Master controller (USB 3.0 Port)

10. Turn on 3 PIN PLUG Switch as below Figure 11.

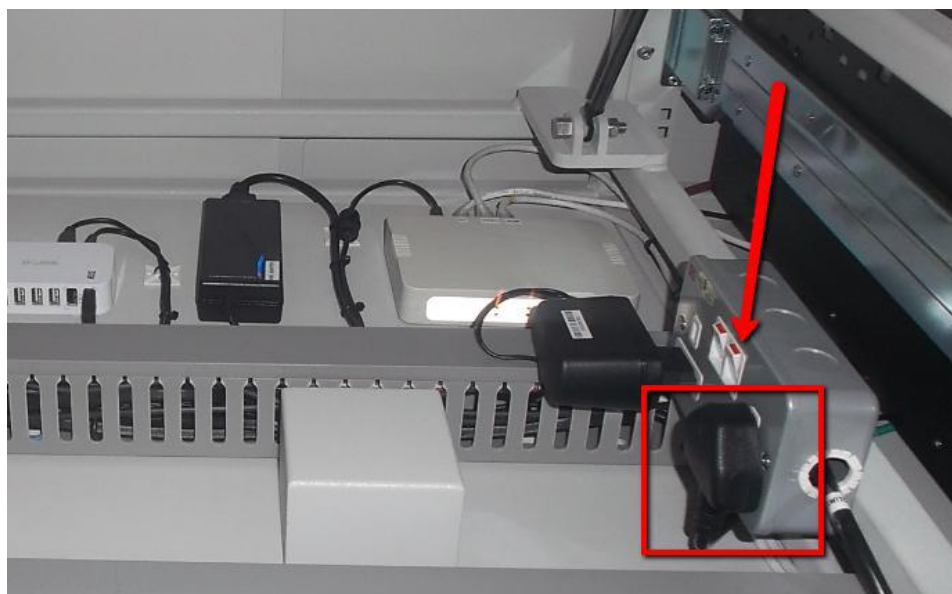


Figure 11 3 PIN PLUG POWER SWITCH

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

11. Go to Control Panel -> Network and Internet -> View Network and tasks as below Figure 12.



Figure 12 Network and Internet

12. Choose "Change Adapter Settings" as below Figure 13.

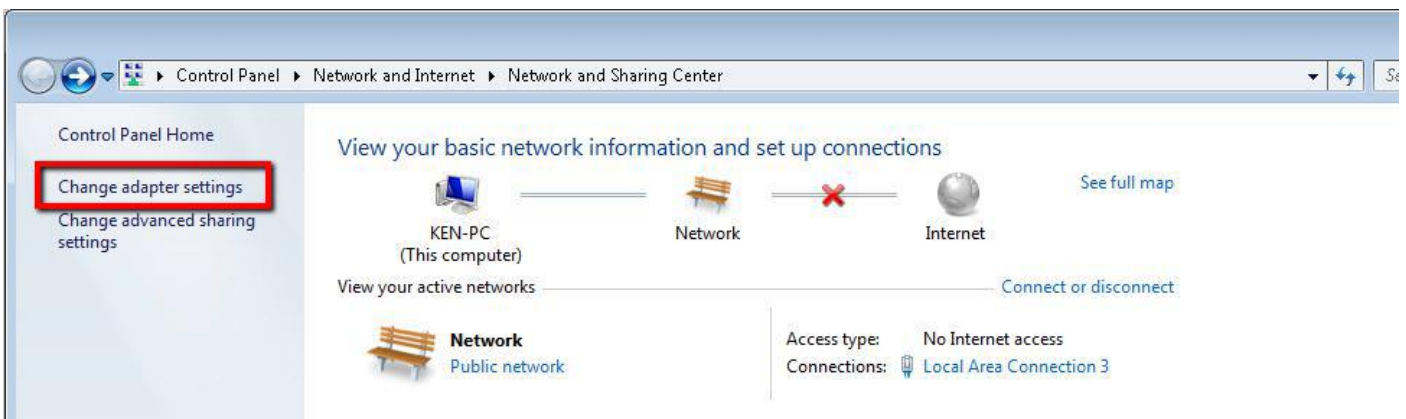


Figure 13 Adapter Setting

13. Choose the "PSP EBLE" as Figure 14.



Figure 14 PSP EBLE Ethernet

14. Right click "PSP EBLE" and select "Properties"

15. Scroll Down and double click “TCP//IPv4” and Set the IP as below Figure 15.

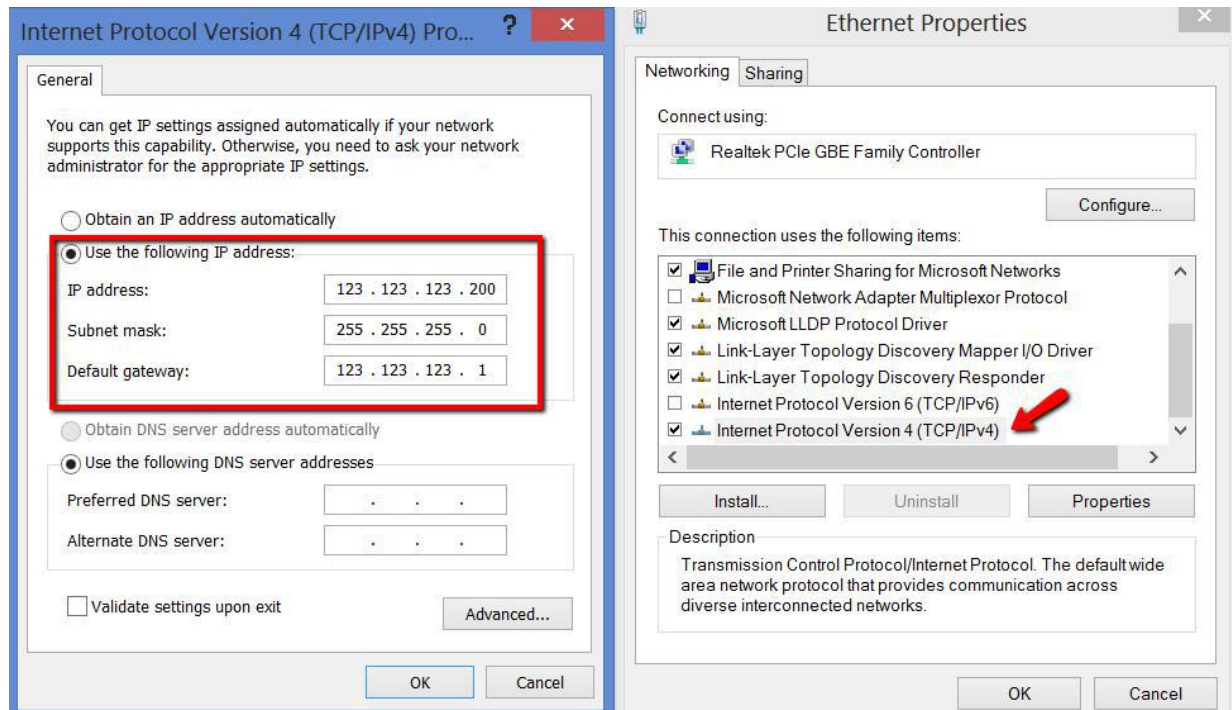


Figure 15 EBLE (PROJECTOR IP PORT) SETTING

16. Perform Ping Test at CMD by using OLD MACHINE IP & NEW MACHINE IP.

May refer to below table:

Type	IP ADDRESS (REAR MODULE)
OLD MACHINE	192.168.0.101
NEW MACHINE	123.123.123.101

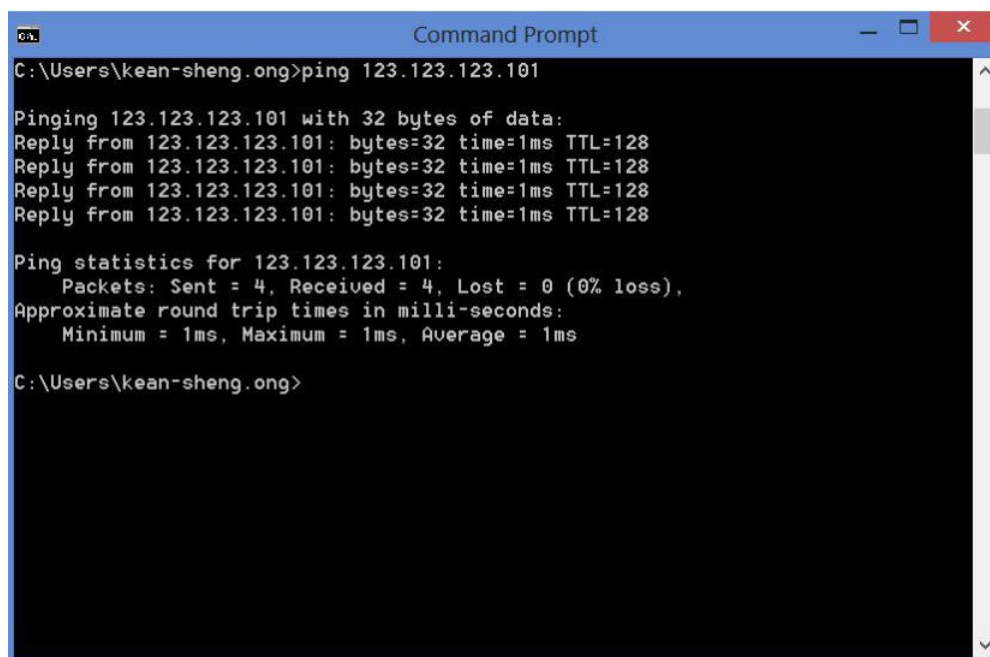
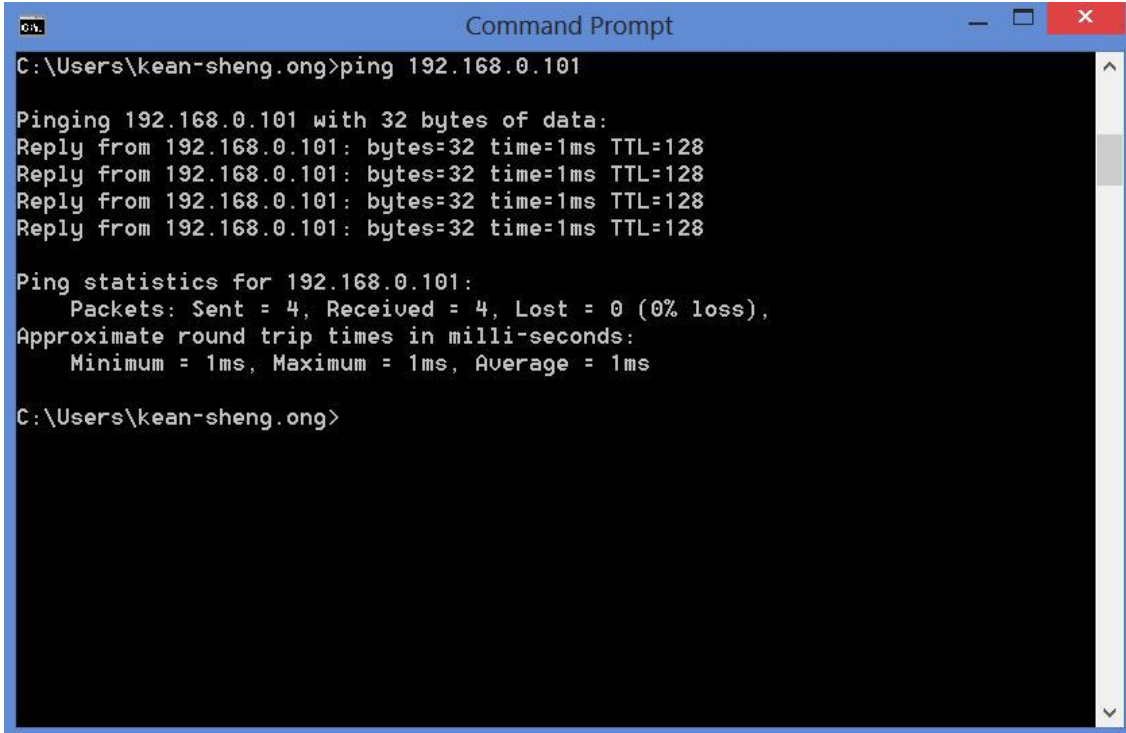


Figure 16 New Rear Module IP Setting (Reference Only)

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016



```

C:\Users\kean-sheng.ong>ping 192.168.0.101

Pinging 192.168.0.101 with 32 bytes of data:
Reply from 192.168.0.101: bytes=32 time=1ms TTL=128
Reply from 192.168.0.101: bytes=32 time=1ms TTL=128
Reply from 192.168.0.101: bytes=32 time=1ms TTL=128
Reply from 192.168.0.101: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.0.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\kean-sheng.ong>

```

Figure 17 Old Rear Module IP Setting (Reference Only)

17. If IP Address “192.168.0.101” do not appear with return value as Figure 17, please refer Step 18-A to Step 18-R for Projector IP Configuration. Once Step 18-A to Step 18-R completed, may continue with Step 19.

If IP Address “123.123.123.101” is appear with return value as Figure 16 and may direct go to Step 19.

18. Perform Projector IP Configuration.

- a. Go to Control Panel -> Network and Internet -> View Network and tasks.
- b. Choose “Change Adapter Settings”
- c. Choose the “PSP EBLE”.
- d. Right click “PSP EBLE” and select properties.

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

e. Scroll Down and double click “TCP//IPv4” and Set the IP as below Figure 18.

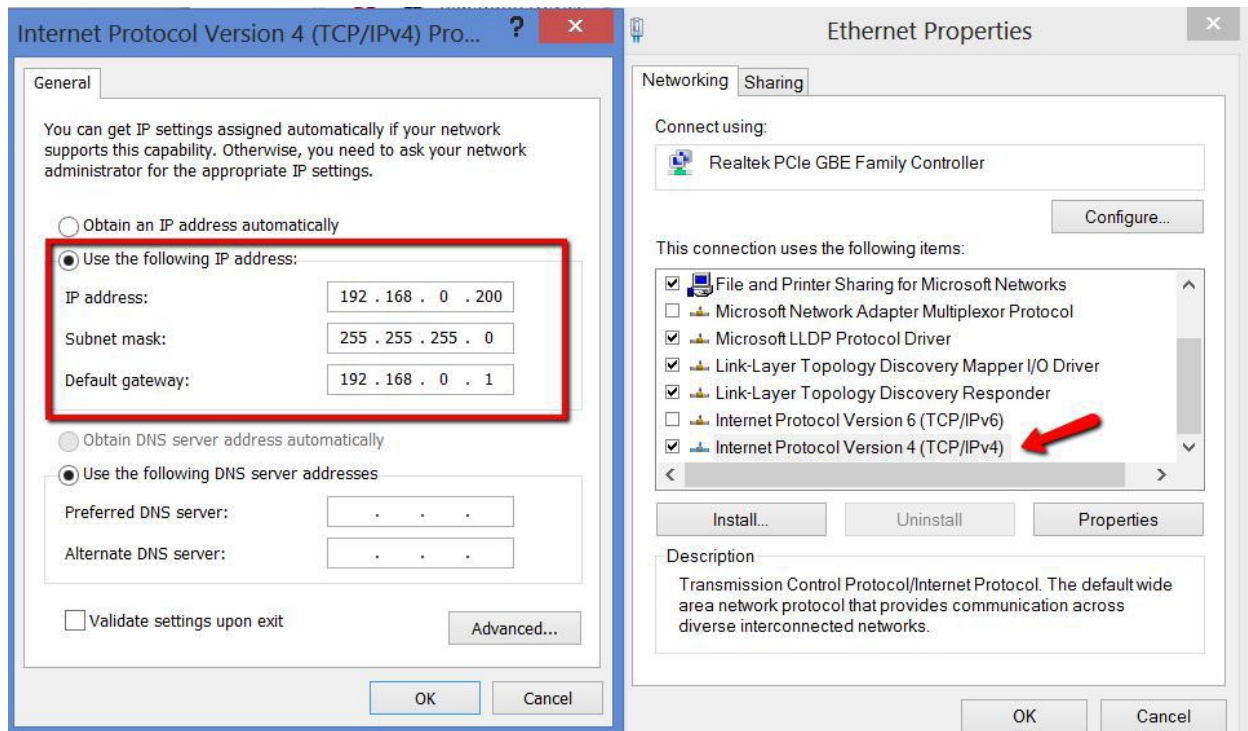


Figure 18 OLD IP SETTING

f. Open up Internet Browser EG: Google Chrome, Internet Explorer or Mozilla Firefox.

g. Type “http://192.168.0.101” as below Figure 19.

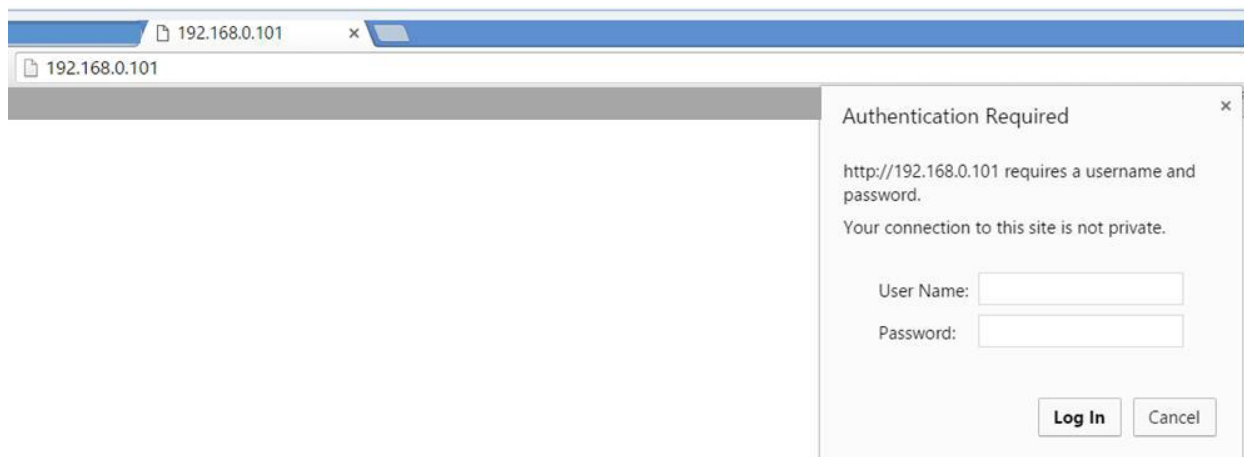


Figure 19 Rear projector IP address

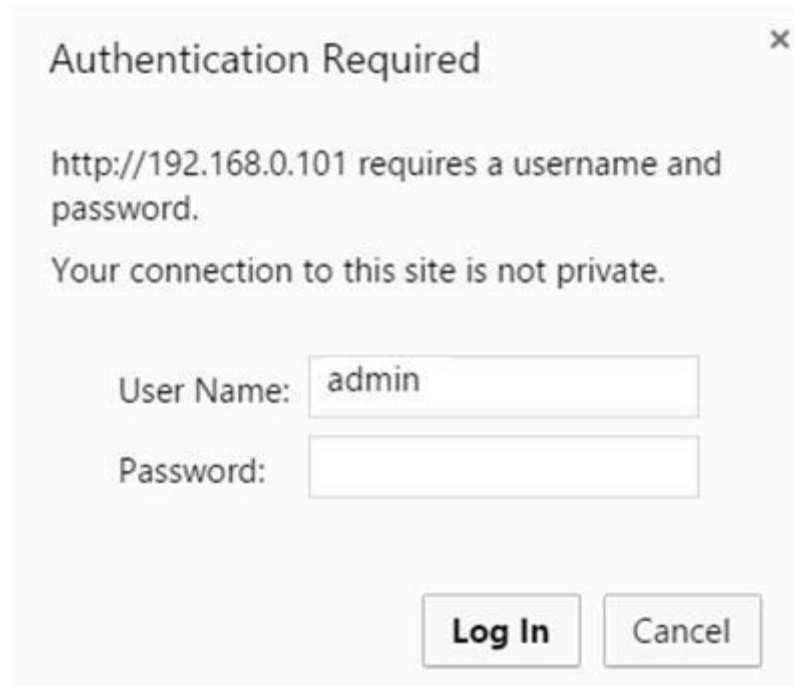
ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

- h. Apply below information at the Authentication Dialog box. Please refer to below Figure 20.

User Name: Admin

Password : <Leave Blank>

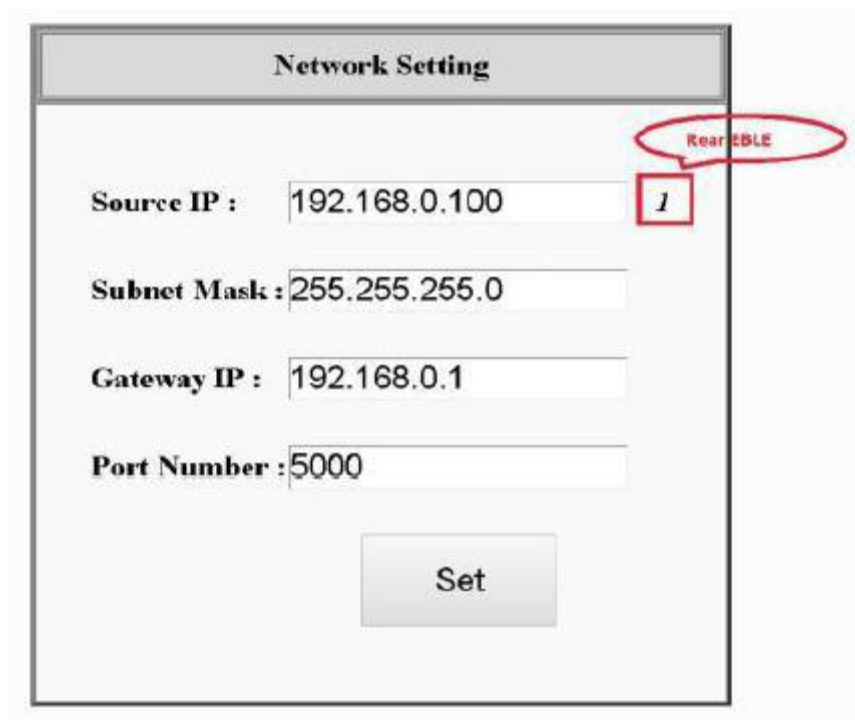
Then, Press “Log In”



The image shows a web browser dialog box titled "Authentication Required" with a close button (X) in the top right corner. The text inside the dialog box reads: "http://192.168.0.101 requires a username and password." followed by "Your connection to this site is not private." Below this text are two input fields: "User Name:" with the text "admin" entered, and "Password:" which is empty. At the bottom right of the dialog box are two buttons: "Log In" and "Cancel".

Figure 20 Authentication Dialog Box

- i. Network Setting dialog box will be displayed as below Figure 21.



The image shows a "Network Setting" dialog box. It has a title bar with the text "Network Setting". Inside the dialog box, there are four labeled input fields: "Source IP :" with the value "192.168.0.100", "Subnet Mask :" with the value "255.255.255.0", "Gateway IP :" with the value "192.168.0.1", and "Port Number :" with the value "5000". Below these fields is a "Set" button. A red oval highlights the text "Rear LED" in red, and a red square highlights the number "1" in the Source IP field.

Figure 21 Rear Projector Network Setting

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

j. Perform IP Configuration by refer to below Figure 22.

Figure 22 Rear Projector New IP Setting

- k. Press “Set” and close Internet Browser
- l. Go to Control Panel -> Network and Internet -> View Network and tasks.
- m. Choose “Change Adapter Settings”
- n. Choose the “ PSP EBLE”
- o. Right click “PSP EBLE” and select “Properties”

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

p. Scroll Down and double click “TCP//IPv4” and Set the IP as below Figure 23.

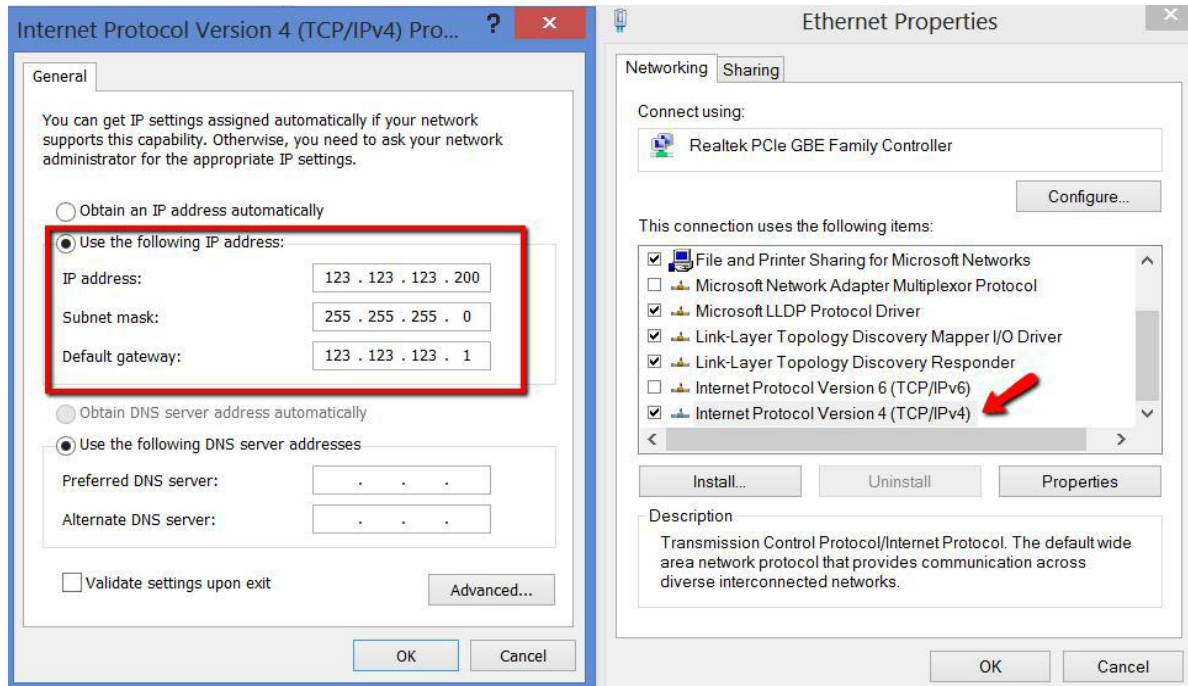


Figure 23 IP Setting

q. Reseat Cat 5 cable at Projector Module.

r. Perform Ping Test at CMD by type “ping 123.123.123.101”

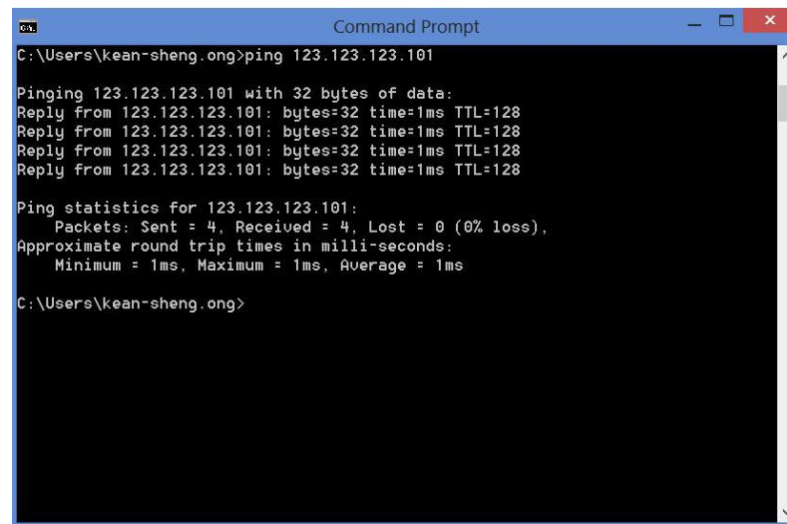


Figure 24 New Projector IP Address

19. Open Hardware Config by refer below path where depend Equipment Model :

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\XXLconfig

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

21. Change the key as below stated :

#####

ethernetBasedLightEngineHostIpAddress=123.123.123.200

ethernetBasedLightEngineCurrentSettingMicroAmps=350

ethernetBasedLightEnginePeriod=20

ethernetBasedLightEngine0IpAddress=123.123.123.100

ethernetBasedLightEngine0Port=5000

ethernetBasedLightEngine1IpAddress=123.123.123.101

ethernetBasedLightEngine1Port=5000

22. Change key at Hardware Config as below table.

System Type	Machine S/N	Adjustment Method	Hardware Config Key Changes
V810 Standard	MY10133110 & below	Manual Load Adjustment Plate	pspModuleVersion=2 pspCalibrationVersion=4 pspSetting=3 opticalCameraModel=axiOpticalCameraType2
V810 Standard	MY10133111 & above	On Rail Adjustment Plate	pspModuleVersion=2 pspCalibrationVersion=3 pspSetting=3 opticalCameraModel=axiOpticalCameraType2
V810 XXL	All	On Rail Adjustment Plate	pspModuleVersion=2 pspCalibrationVersion=3 pspSetting=3 opticalCameraModel=axiOpticalCameraType2

23. Open Software Config by refer to below path where depends on Equipment Model :

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\Config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\xxlConfig

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

24. Change the key as below Figure 25:

```
# online or offline system
onlineWorkstation=true

# All hardware subsystems simulation
# Note: this parameter will override all of the individual hardware subsystems when set to
"true"
hardwareSimulation=false
```

Figure 25 Software Config

25. Identify Camera Orientation for Rear PSP 2 Module as below Figure 26.



Figure 26 Rear PSP 2 Module (Camera Orientation)

26. Open StCamWare at desktop as below Figure 27.



Figure 27 StCamsWare

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

27. Optical Camera Live View Dialog box would be illustrated as below Figure 28.

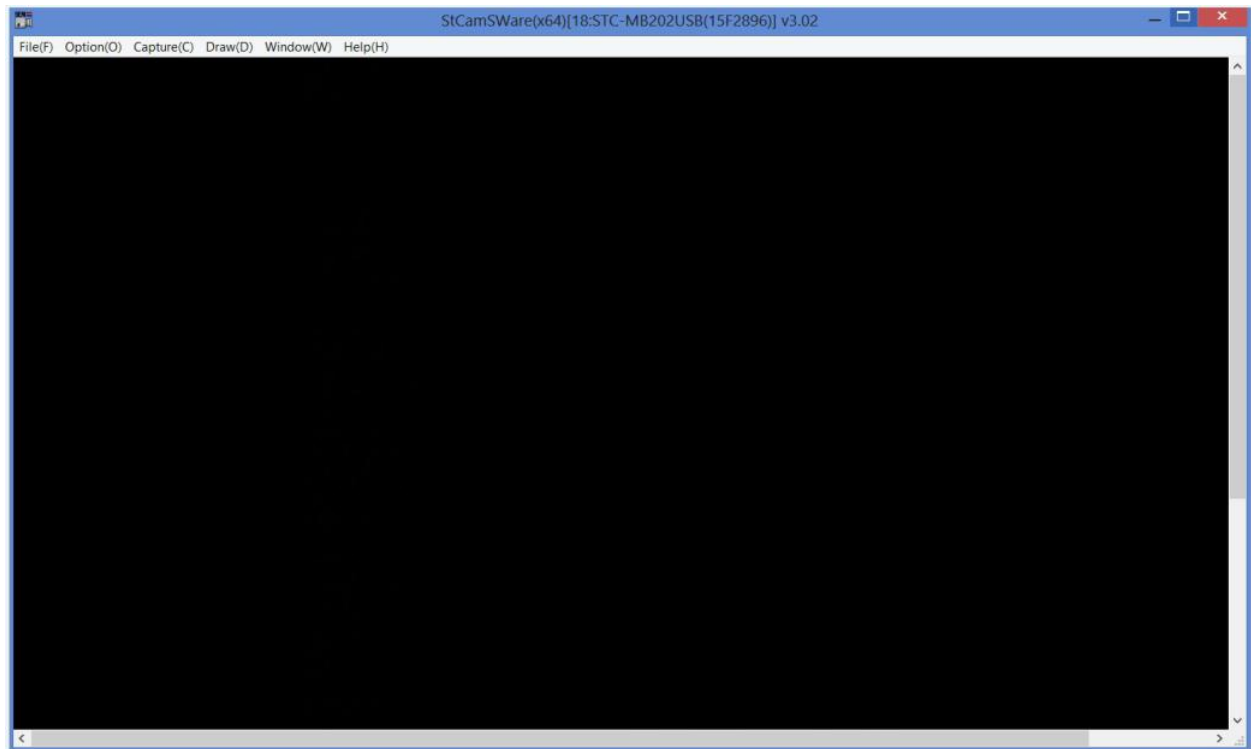


Figure 28 Optical Camera Live View

28. Go to "Option" and select Rename Camera (R) as below Figure 29.

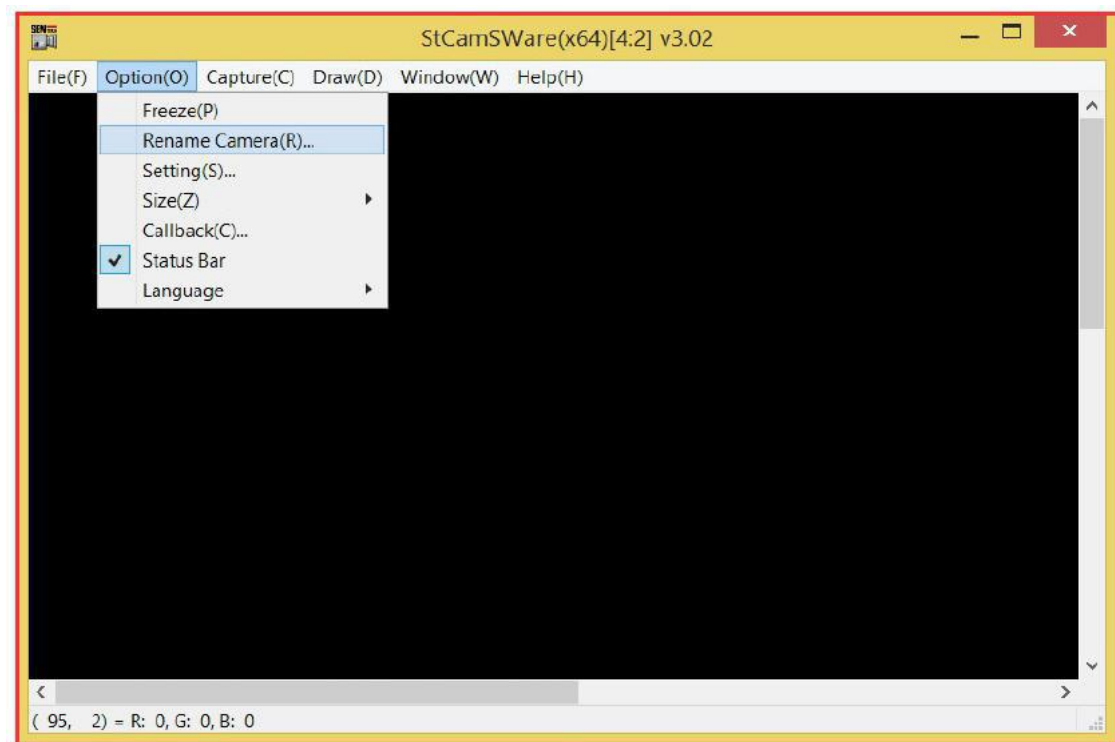


Figure 29 Optical Camera Live View

29. Camera Name Dialog box would be illustrated as below Figure 30.

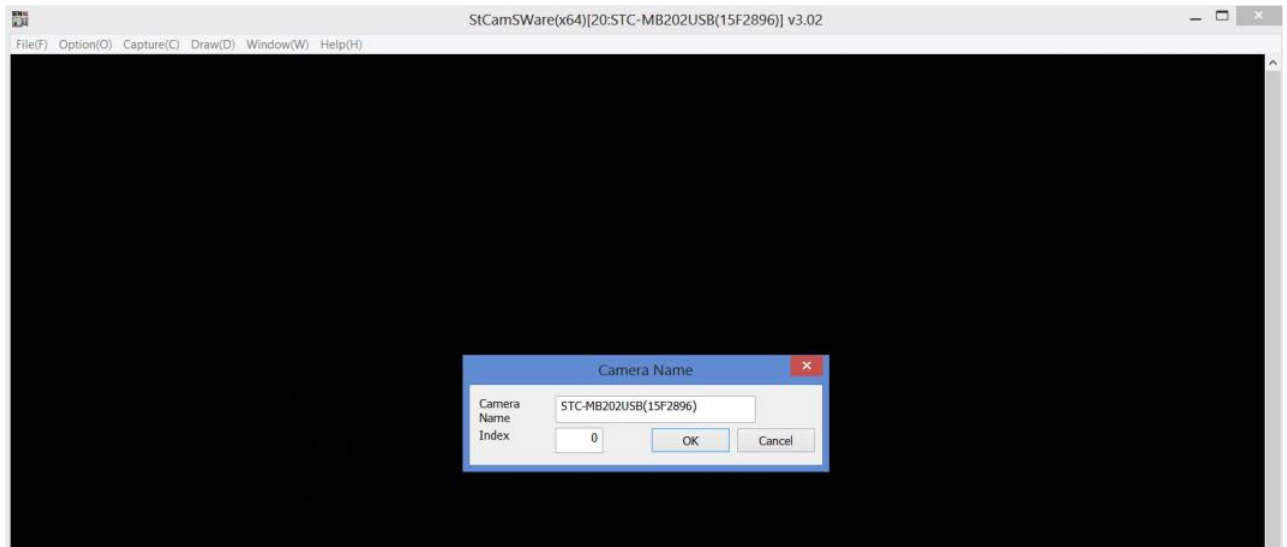


Figure 30 Camera Name Dialog Box

30. Configure Rear PSP 2 Camera Module by refer to below Figure 31 and Click "OK".

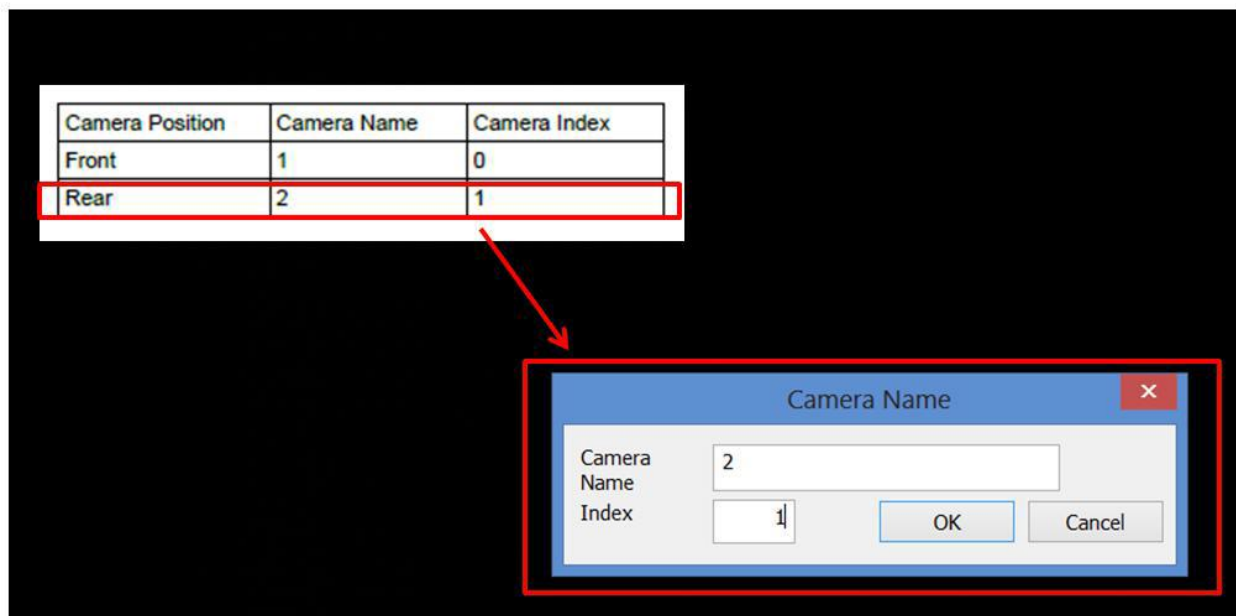


Figure 31 Camera Name Dialog box

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

31. Open V810 Software & go to Services-> Subsystem Status& Control-> Ethernet-Based Light Engine and press Initialized All EBLEs as below Figure 32.

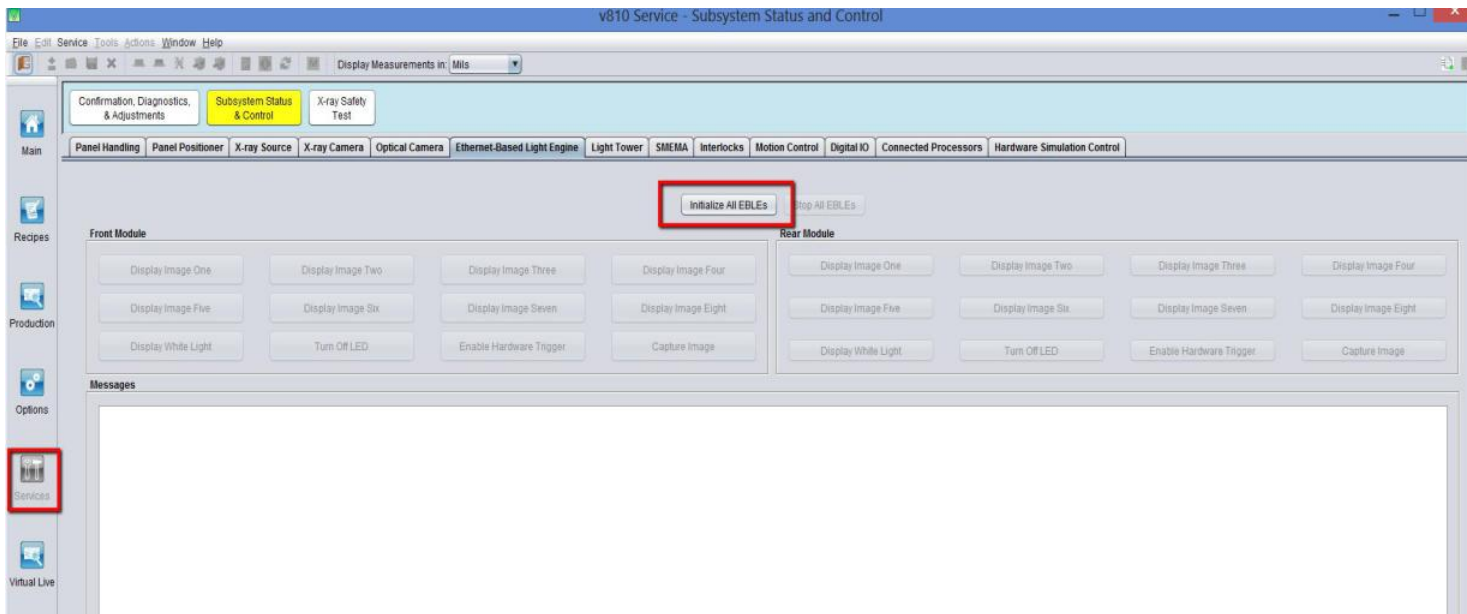


Figure 32 EBLE GUI

32. At Rear Module Column, please click “Display White Light” as below Figure 33.



Figure 33 Rear Module Column

33. At StCamsWare, Go to Option and Select Setting as below Figure 34.

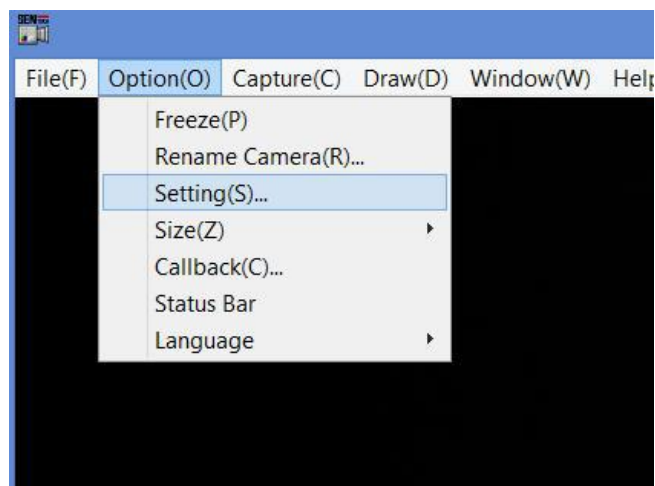


Figure 34 StCamsWare GUI

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

34. Select “OFF (Normal)” at Mirror Image Column as below Figure 35.

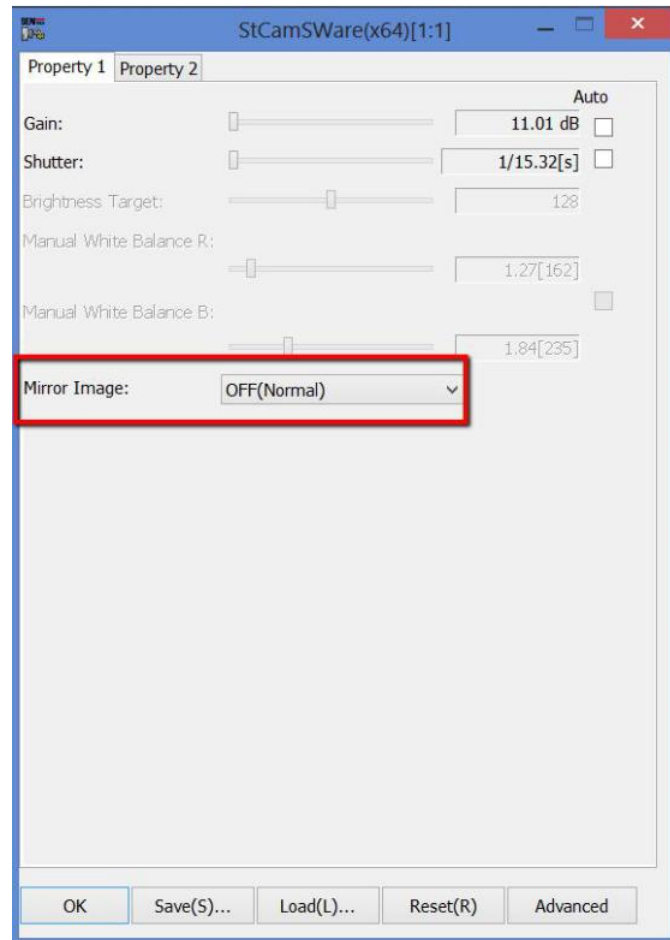


Figure 35 StCamsWare GUI

35. Loosen 2 x Screws and extend the module towards Front Direction as below Figure 36. Tighten 2 x Screws once module is fully extended.

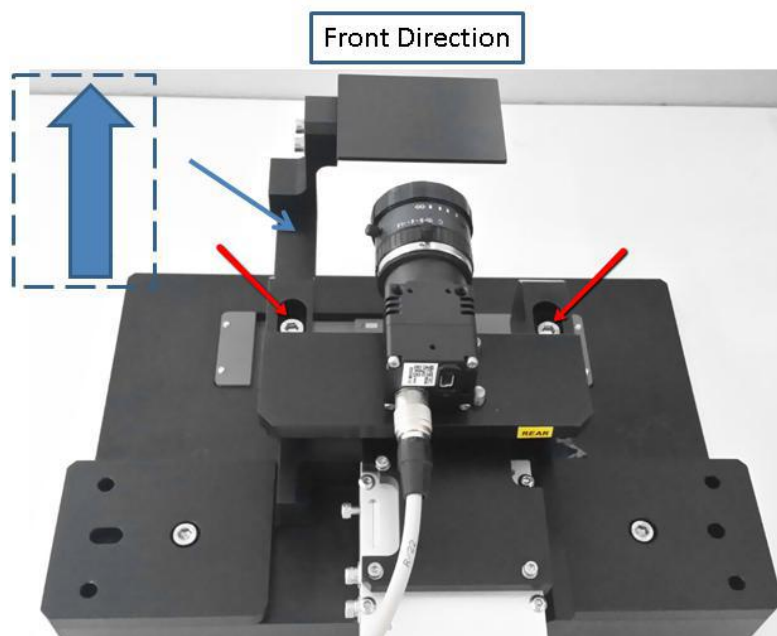


Figure 36 Module Extend towards Front Direction

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

36. Loosen 2 x Side Screws of Projector Module as below Figure 37.

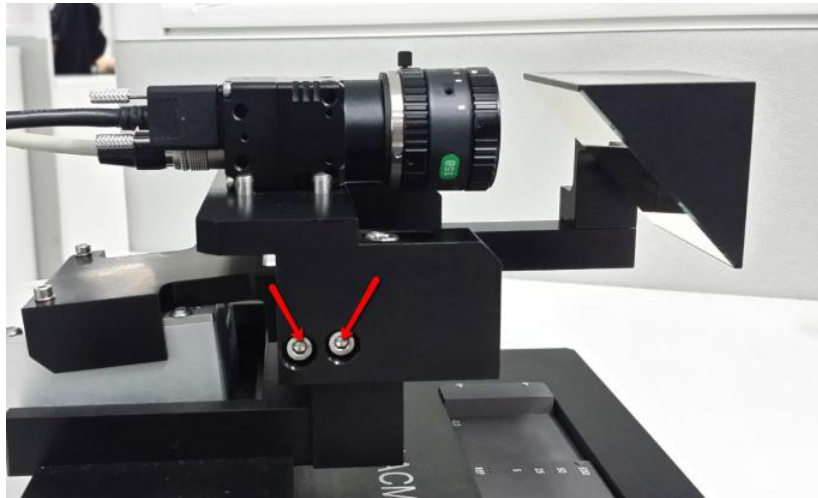


Figure 37 Projector Module Side Screws

37. Adjust the Projector angle until white light projection is fully covered on the adjustment plate as below Figure 38.

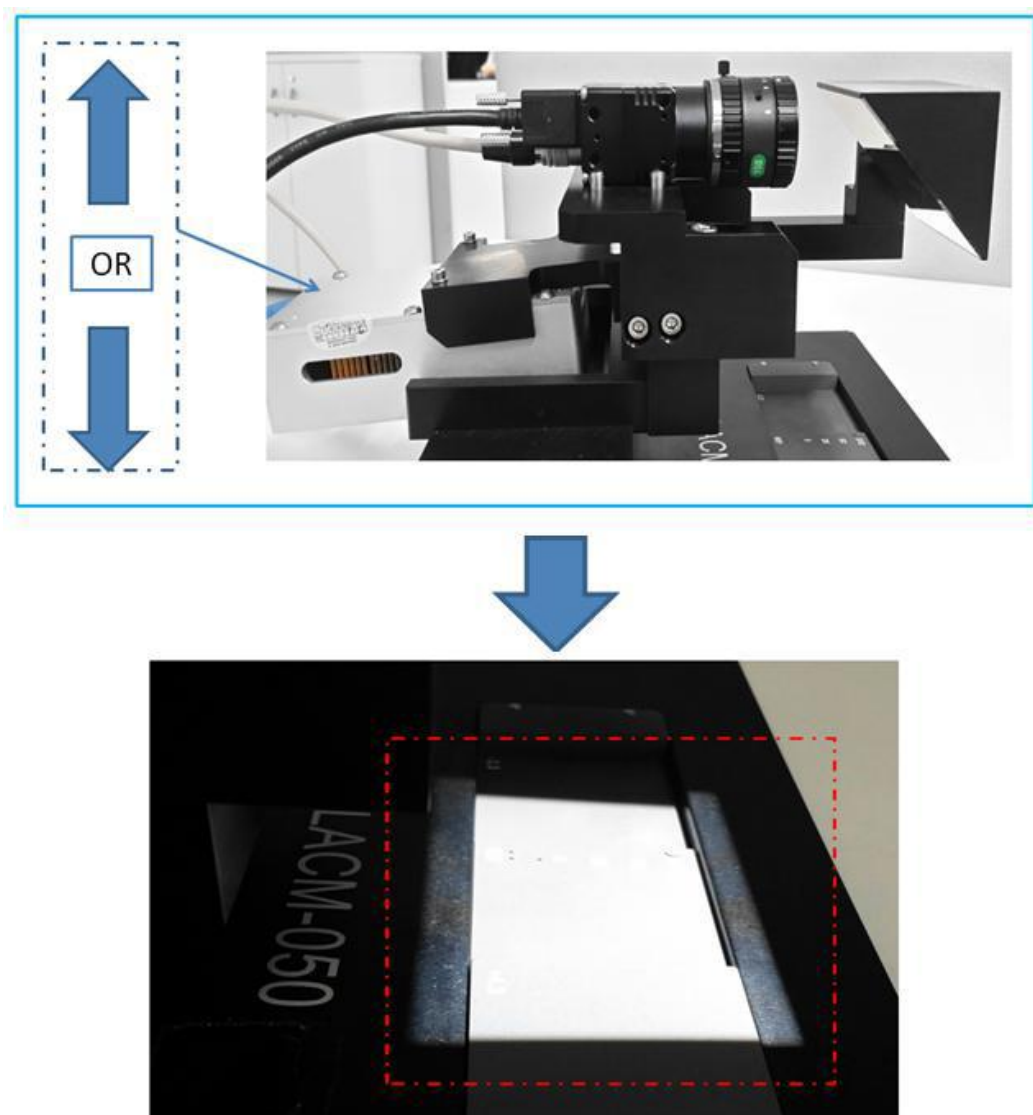


Figure 38 White Light Projection Coverage Area

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

38. Go to StCamSware, go to Option -> Size, Select "x 1" as below Figure 39.

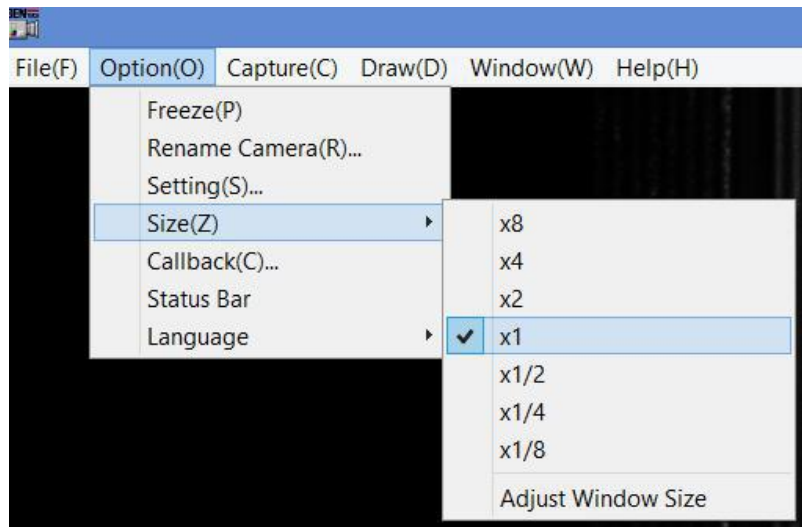


Figure 39 StcamSware GUI

39. Open StCamsWare and verify the white light projection image is within the desired coverage area as below Figure 40.

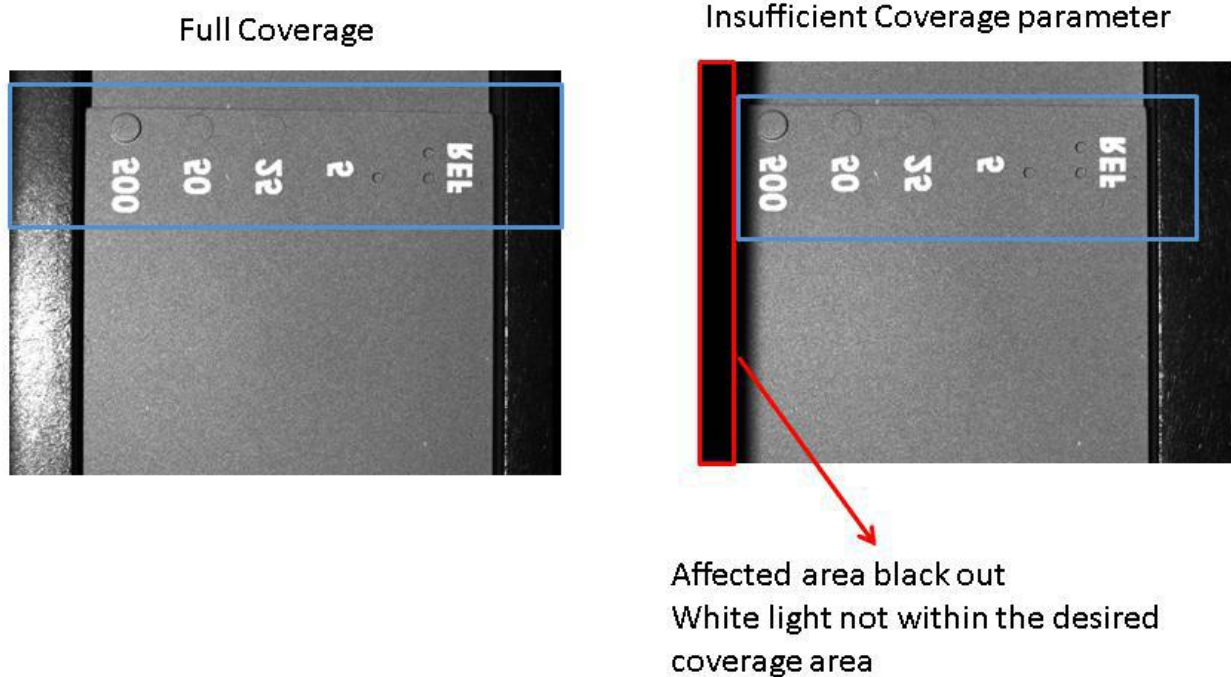


Figure 40 White Light Coverage area

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

40. Next, Go to StCamSware, go to Option -> Size, Select “x ½” as below Figure 41.

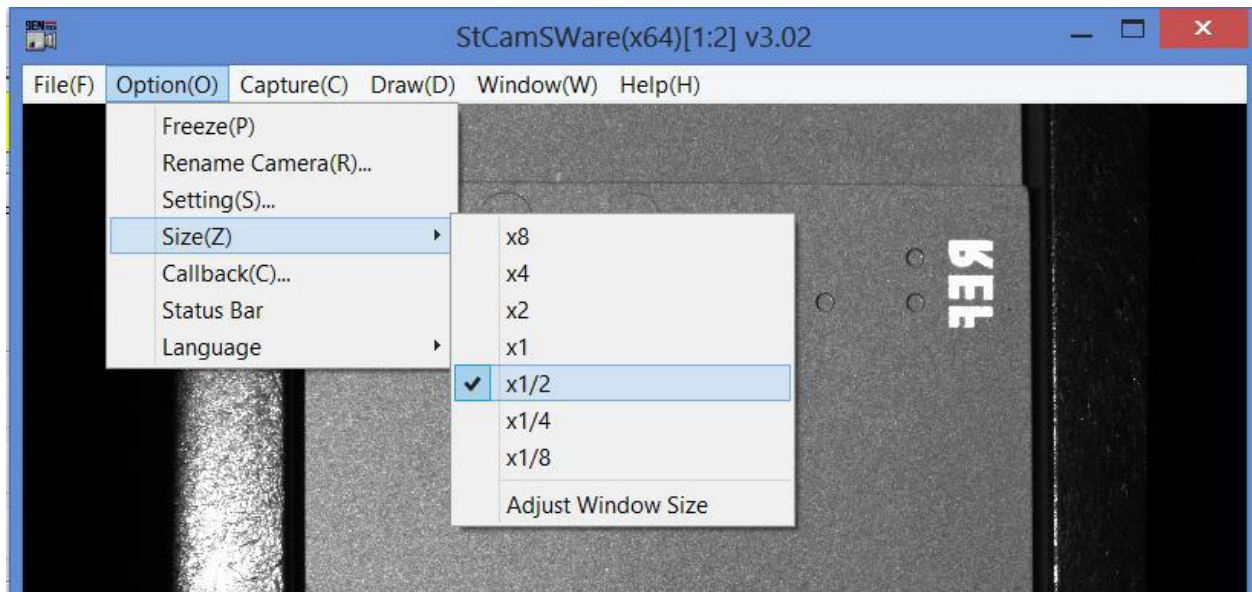


Figure 41 StCamSware Frame Size Setting

41. At EBLE tab, Trigger “Display White Light” at Rear Module Column.

Remarks: If white light had been triggered previously, may ignore this Step.

42. Loosen the “Tiny knob” at the optical camera as below Figure 42.



Figure 42 Optical camera

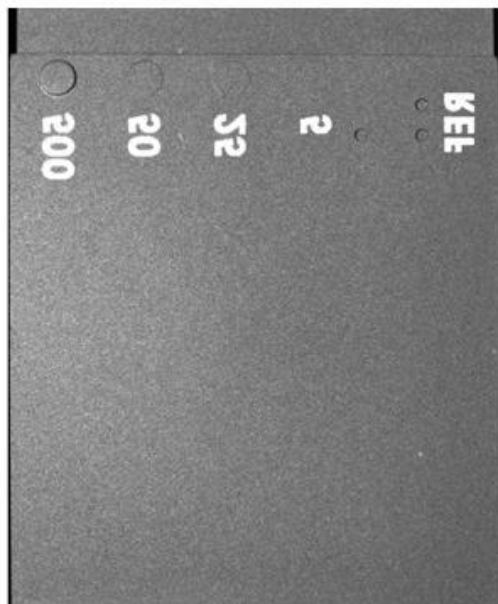
	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

43. Adjust Optical Camera Focus by turning the dial of optical camera as below Figure 43.

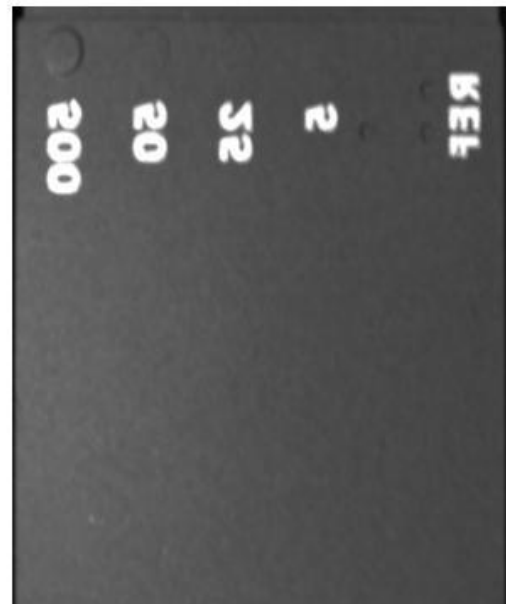


Figure 43 Optical Camera Dial Turning

44. Verify wording on all plane in focus. (REF plane, 5 plane, 25 Plane, 50 plane, 500 plane) as below Figure 44. Repeat Step 42 to 43 until all are plane in focus. Tighten the dial of optical camera once completed.



Optical camera is focus.



Optical camera is NOT focus.

Figure 44 All Plane Focusing

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

45. Next, Go to EBLE tab at V810 Software and Select “Display Image Two” as below Figure 45.



Figure 45 EBLE GUI

46. Fringe pattern would illustrate at StCamSware GUI as below Figure 46.



Figure 46 Fringe Pattern

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

47. Go to StCamSware GUI ->Option and Select Size (Z) “x1” as below Figure 47.

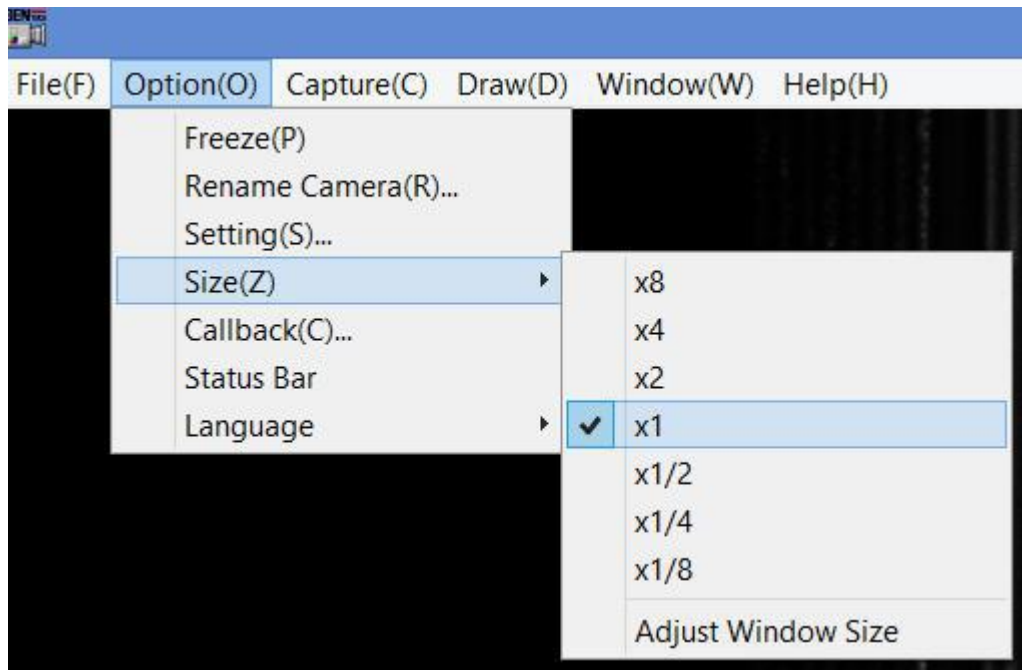


Figure 47 StCamSware GUI

48. Loosen 2 X Screws at EBLE Projector as below Figure 48.

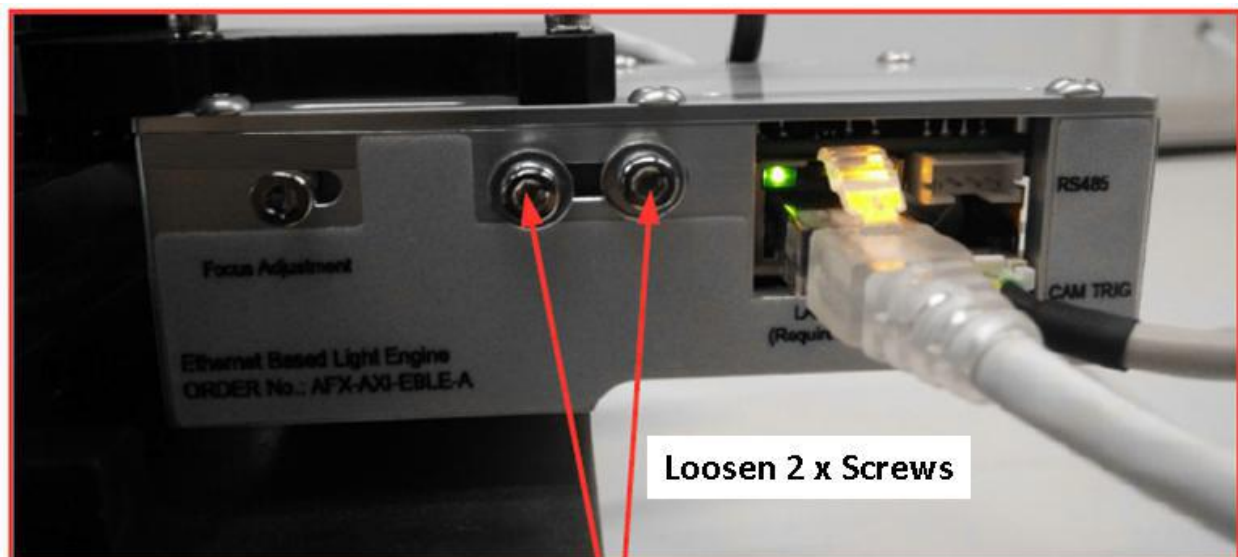


Figure 48 EBLE Projector

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

49. Adjust the Focus Adjustment until “Clear Fringe Pattern” is obtained as Figure 49. Then, Tighten 2 x Screws.

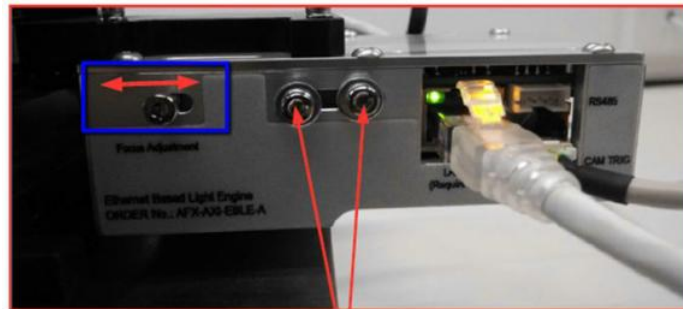


Figure 49 Clear Fringe Pattern

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

50. Then, Go to StCamSware GUI ->Option and Select Size (Z) “x 8” as below Figure 50.

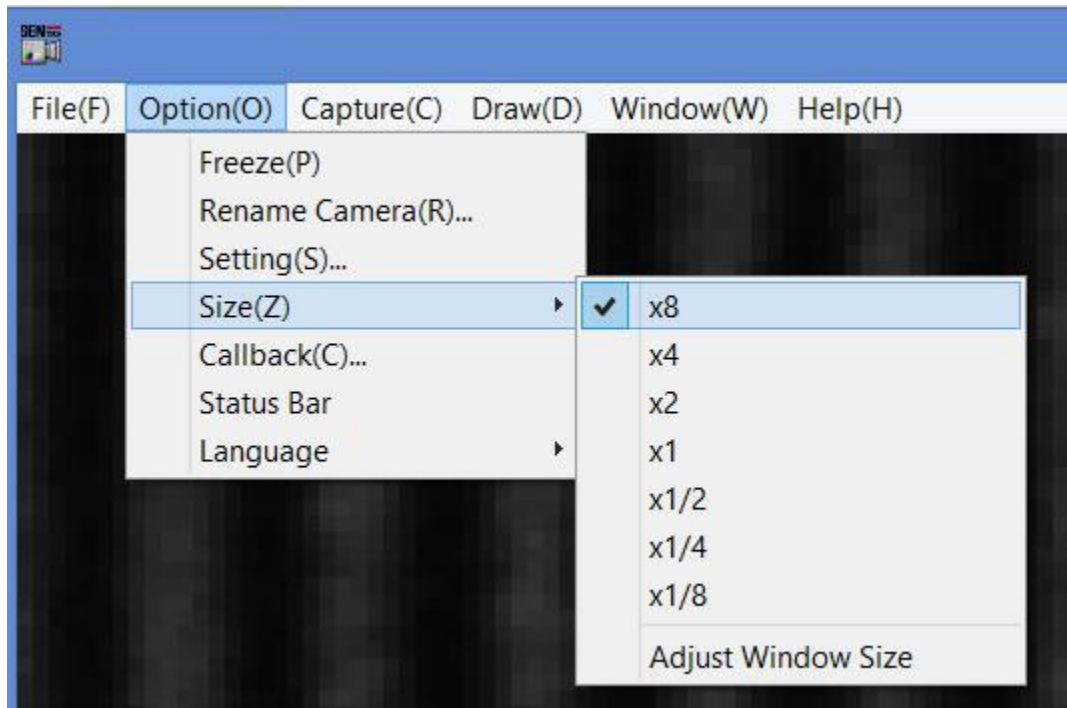


Figure 50 StcamSware GUI

51. Verify the Fringe pattern image sharpness are approximately similar to Image A (Figure 51)

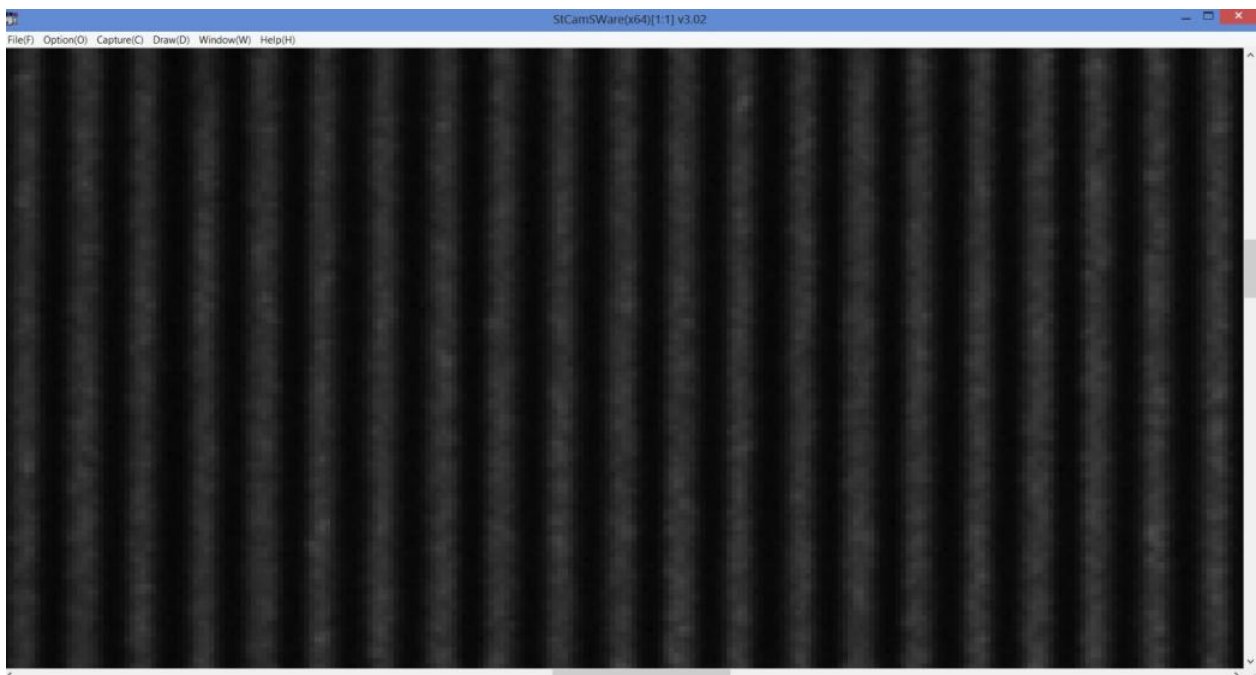


Figure 51 Image A

a.If Image A is obtained during the verification, may direct proceed to Step 51.

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

- b. If Image B is obtained during the verification, please re-adjust the “Focus adjustment” of projector by refer to “Size 8” at Stcamsware GUI until Image A (Figure 51) is obtained. Once completed, please re-verify the sharpness image at “Size 1” at Stcamsware GUI as Figure 49.

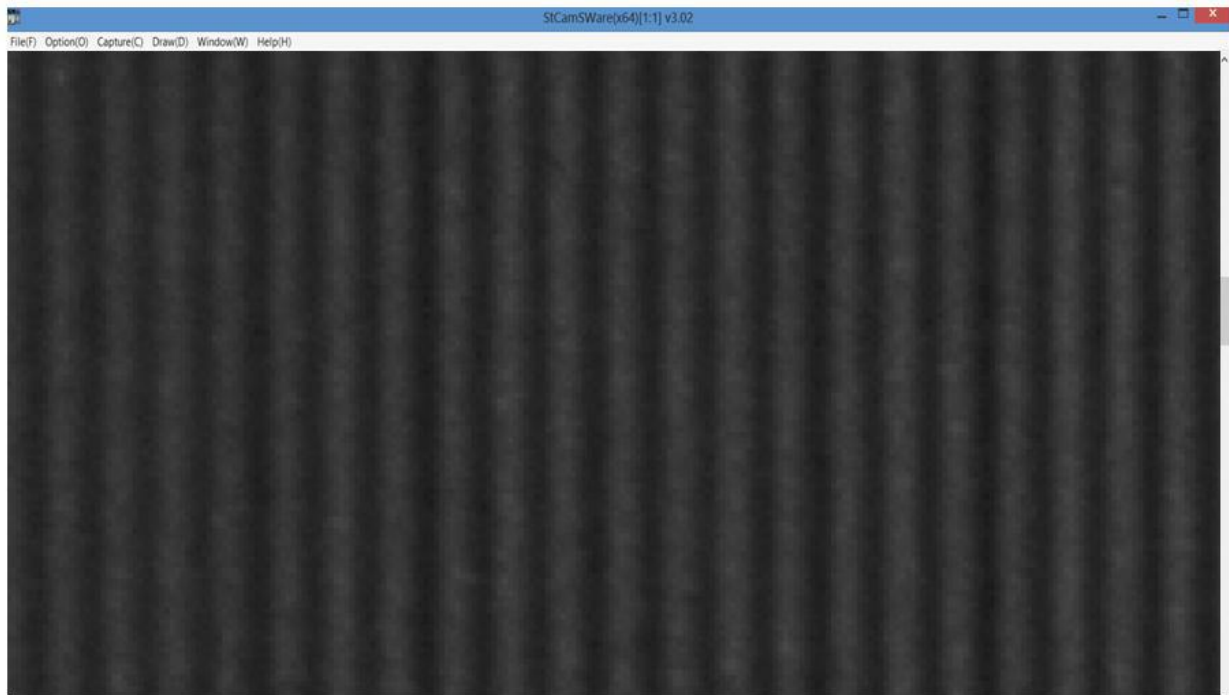


Figure 52 Image B

52. Next, Close the StcamSware Software.

53. Obtain “Aperture_StCamWare” from AXI HW Support Team. Figure 53 as below.



Figure 53 Aperture_StCamWare

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

54. Double Click “Aperture_StCamWare” and 3 file will be displayed as below Figure 54.

 StCtlLib.dll	2/4/2015 10:17 AM	Application extens...	14 KB
 StTrgDisplayVCS2005.exe_S2EX	7/31/2015 5:30 PM	EXE_S2EX File	192 KB
 StTrgDisplayVCS2005.exe_StdXXL	7/27/2015 10:56 A...	EXE_STDXXL File	192 KB

Figure 54 Aperture_StCamWare File Type

55. Rename the “**StTrgDisplayVCS2005.exe_StdXXL**”to“**StTrgDisplayVCS2005.exe**” as below Figure 55.

 StCtlLib.dll	2/4/2015 10:17 AM	Application extens...	14 KB
 StTrgDisplayVCS2005.exe_S2EX	7/31/2015 5:30 PM	EXE_S2EX File	192 KB
 StTrgDisplayVCS2005.exe_StdXXL	7/27/2015 10:56 A...	EXE_STDXXL File	192 KB



 StCtlLib.dll	2/4/2015 10:17 AM	Application extens...	14 KB
 StTrgDisplayVCS2005.exe	7/27/2015 10:56 A...	Application	192 KB
 StTrgDisplayVCS2005.exe_S2EX	7/31/2015 5:30 PM	EXE_S2EX File	192 KB

Figure 55 Exe File rename

56. Then, Go to EBLE tab at V810 Software and select “Display White Light “.

57. Double click the new rename file (**StTrgDisplayVCS2005.exe**) where under Aperture_StCamWare Folder.

58. A Dialog box with live View will be illustrated as below Figure 56.

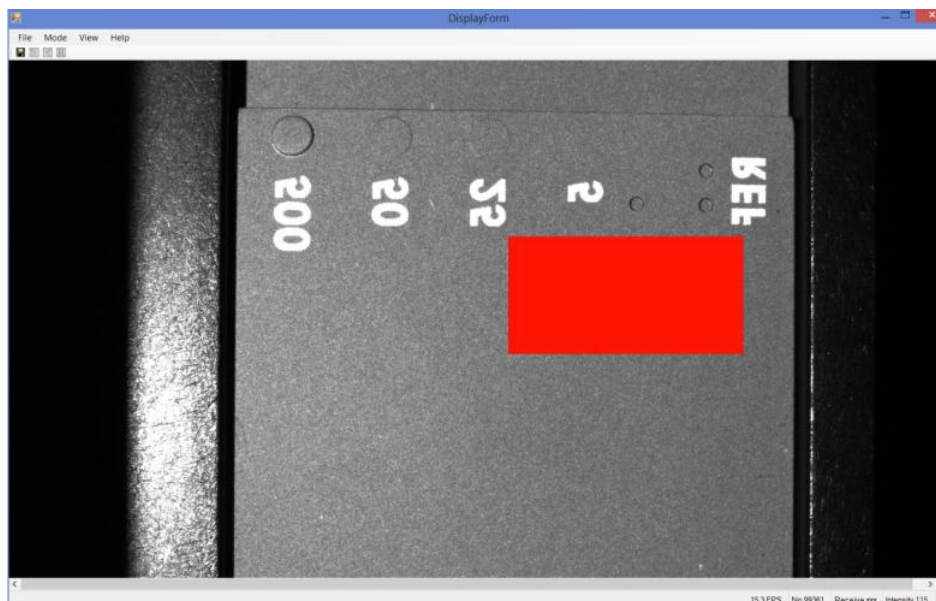


Figure 56 Display Form Dialog Box

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

59. Loosen the “Tiny Knob” at Optical Camera as below Figure 57.



Figure 57 Tiny Knob at Optical Camera

60. Turning the dial for Optical Camera Aperture as below Figure 58.

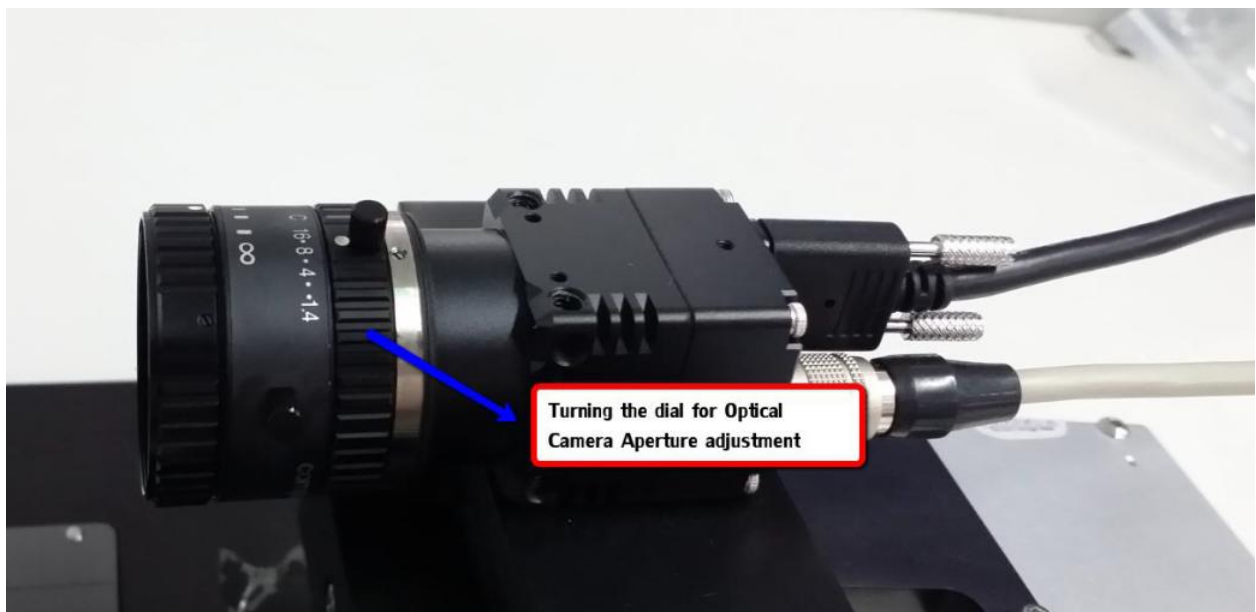
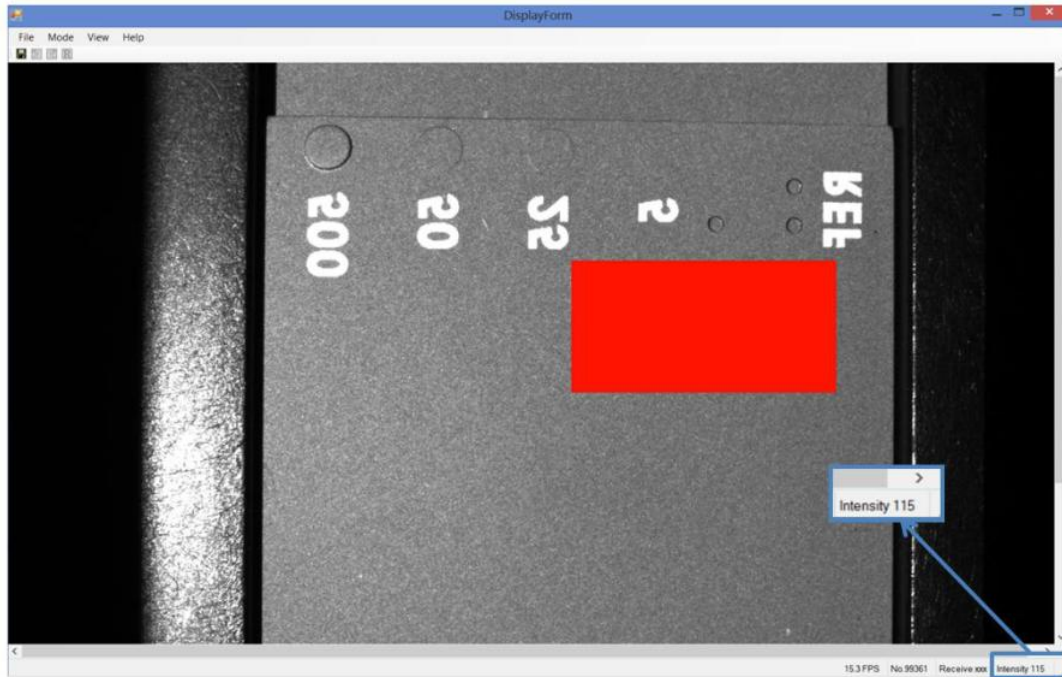


Figure 58 Aperture Dial Turning

	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

61. Stop the turning when Intensity is within below range as below Figure 59. Once complete, tighten the “Tiny Knob” where previous mentioned at Step 58.

Intensity range	132-138
-----------------	---------



Poor Adjustment



Well Adjustment

Figure 59 Intensity range

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 PSP Rear Module Offline Adjustment Procedure (EQUIPMENT)	Effective Date	2 Oct 2016

62. Click “ Stop All EBLE” at EBLE Tab (V810 Software)
63. Close the “Display Form Dialog Box” as Figure 56.
64. Under “Aperture_StCamWare” Folder and rename the
“StTrgDisplayVCS2005.exe”to“StTrgDisplayVCS2005.exe_StdXXL” as below Figure 60.

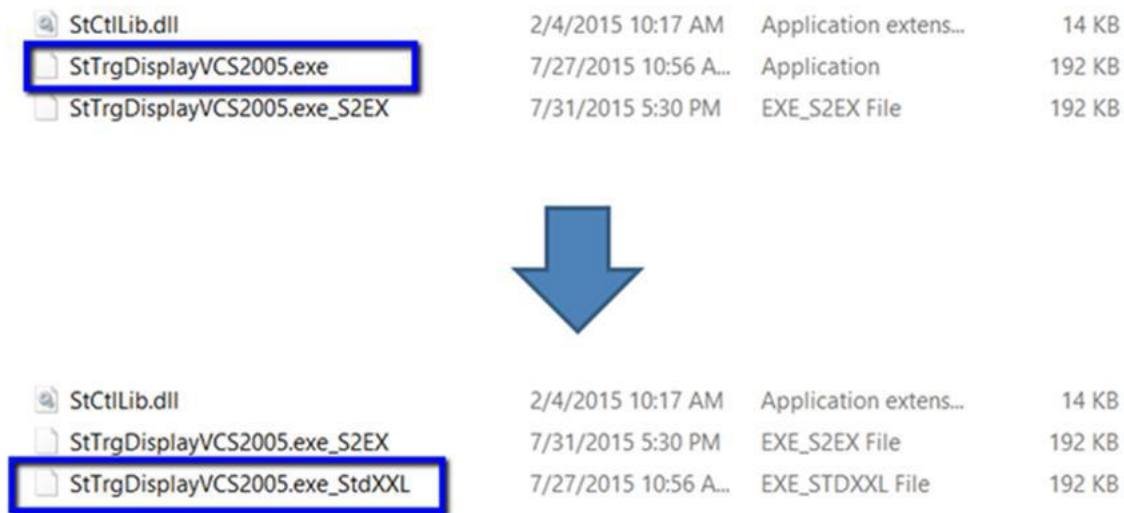


Figure 60 Exe File rename

65. Unplug 1 x Blue Cat 5 Cable from PSP 2 Module.
66. Revert to Origin Cat 5 Cable at POE Switch
67. Remove the Optical Camera “Area Scan” Cable from PSP 2 Module.
68. Remove the entire Module and Offline Adjustment Jig from the machine.
69. Exit V810 Software
70. Go to hardware config and change as below table.

System Type	Machine S/N	Adjustment Method	Hardware Config Key Changes
V810 Standard	MY10133110 & below	Manual Load Adjustment Plate	pspSetting=1
V810 Standard	MY10133111 & above	On Rail Adjustment Plate	pspSetting=1
V810 XXL	All	On Rail Adjustment Plate	pspSetting=1

71. Re-open V810 Software.